CIS 613: Exploratory Testing

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How can I get some free McDonalds?

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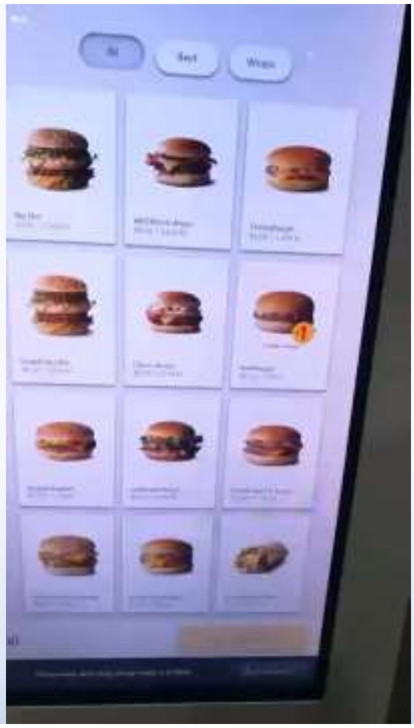

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How can I get some free McDonalds?

- Which, if any, of the testing methods that we have covered could have prevented this?
  - Identify test method
  - Identify steps you would take to find the issue / test

- What is the intrinsic difficulty in identifying this negative behavior?

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What is exploratory testing?

- Explore: to investigate the unknown
- “The opposite of scripted testing”—James Bach
- More than ad hoc testing
- Shares some pros and cons with Special Value Testing
- Depends on the abilities of the exploratory tester
  - Curiosity
  - Ingenuity
  - Time and tools
- A discovery process
- A learning process in which the results of one test suggest additional tests

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James Bach’s Exploration Mindset

“I run a test.

I MAKE AN OBSERVATION...

I experience surprise associated with a pattern within an observation.

That triggers REFLECTION about PLAUSIBILITY...

The pattern seems implausible relative to my current model of the phenomenon.

That triggers REFLECTION about RISK...

I can bring to mind a risk associated with that pattern.

That triggers REFLECTION on MAGNITUDE OF RISK...

The risk seems important.

That triggers TEST REDESIGN... ”

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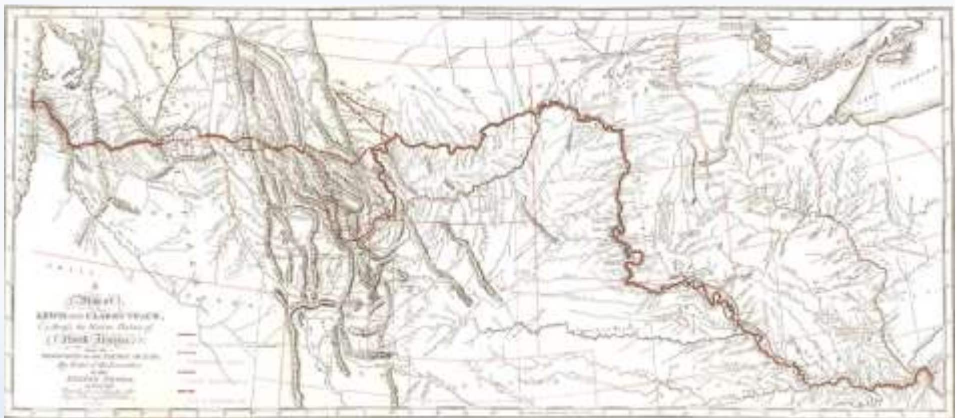
Lewis and Clark Expedition



Purpose: “the most direct and practicable water communication across this continent, for the purposes of commerce” – Thomas Jefferson

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Testing Excellence

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Ingredients of Software Testing Excellence?

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Ingredients of Software Testing Excellence

- Craftsmanship
- Best practices
- “Silver Bullets” for software testing
- Top 10 best testing practices
- Mapping top 10 practices to application types

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Elements of Craftsmanship

- Mastery of
  - the subject matter
  - the associated tools
  - the associated techniques
- The ability to make appropriate choices
  - about tools
  - techniques
- Extensive experience with the subject matter
- A significant history of high quality work with the subject matter
- Pride in one's work

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About "Best Practices"

- Some experts deny that they exist

- Others agree on these attributes
  - They are usually defined by practitioners
  - They are "tried and true"
  - They are very dependent on the subject matter
  - They have a significant history of success

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Top 10 Best Practices for Software Testing Excellence

1. Model-Driven Agile Development
2. Careful Definition and Identification of Levels of Testing
3. System-Level Model-Based Testing
4. System Testing Extensions
5. Incidence Matrices to Guide Regression Testing
6. Use of MM-Paths for Integration Testing
7. Intelligent Combination of Specification-Based and Code-Based Unit Level Testing
8. Code Coverage Metrics Based on the Nature of Individual Units
9. Exploratory Testing During Maintenance
10. Test-Driven Development

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Mapping Best Practices to Application Types

Best Practice	Mission Critical	Time Critical	Legacy Code
Model-Driven Development	x		
Careful Definition and Identification of Levels of Testing	x	x	x
System-Level Model-Based Testing	x		
System Testing Extensions	x		
Use of MM-Paths for Integration Testing	x		
Intelligent Combination of Specification-Based and Code-Based Unit Level Testing	x		x
Code Coverage Metrics Based on the Nature of Individual Units	x		
Exploratory Testing During Maintenance			x
Test-Driven Development		x	

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No Silver Bullet: Essence and Accident in Software Engineering

Written by Fred Brooks in 1986:

- Managed the development of IBM’s System/360 and OS/360 Operating Systems
- Also wrote the “Mythical Man Month” around the theme “adding manpower to a late software project makes it later”
- Won the “National Medal of Technology” in 1985
- Won the “Turing Award” in 1999

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Essential Difficulties

- Complexity
- Conformity
- Changeability
- Invisibility

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Essential Difficulties

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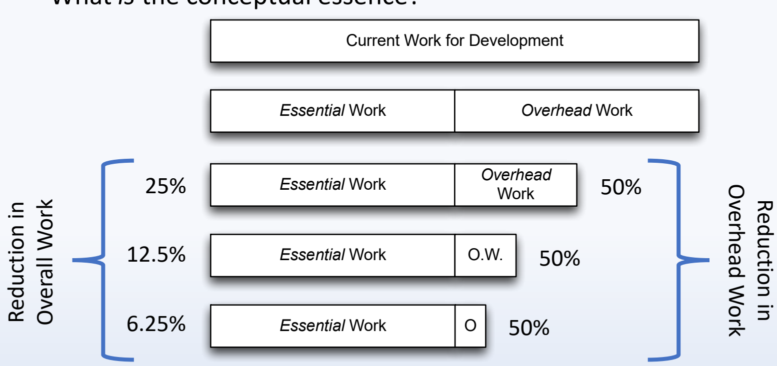
Promising Attacks on the Conceptual Essence

What *is* the conceptual essence?



What would make this conceptual essence invalid?

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Promising Attacks on the Conceptual Essence

- Buy vs Build
- Requirements refinement and rapid prototyping
- Incremental development – grow, not build, software
- Great designers

Any other promising attacks that should be included?

David Harel's answer? "Biting the Silver Bullet" (Do the work!)

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Test Maturity Model

- A Spin-off of Capability Maturity Model
- 5 Levels
  - **Initial**: ad hoc, no quality standard
  - **Definition**: possible defined test process. Testing begins when project development ends.
  - **Integration**: testing integrated with development. Possible independent test teams.
  - **Management and measurement**: (name says most of it), quality criteria in place
  - **Optimization**: tools incorporated into process, emphasis shifts to defect prevention

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